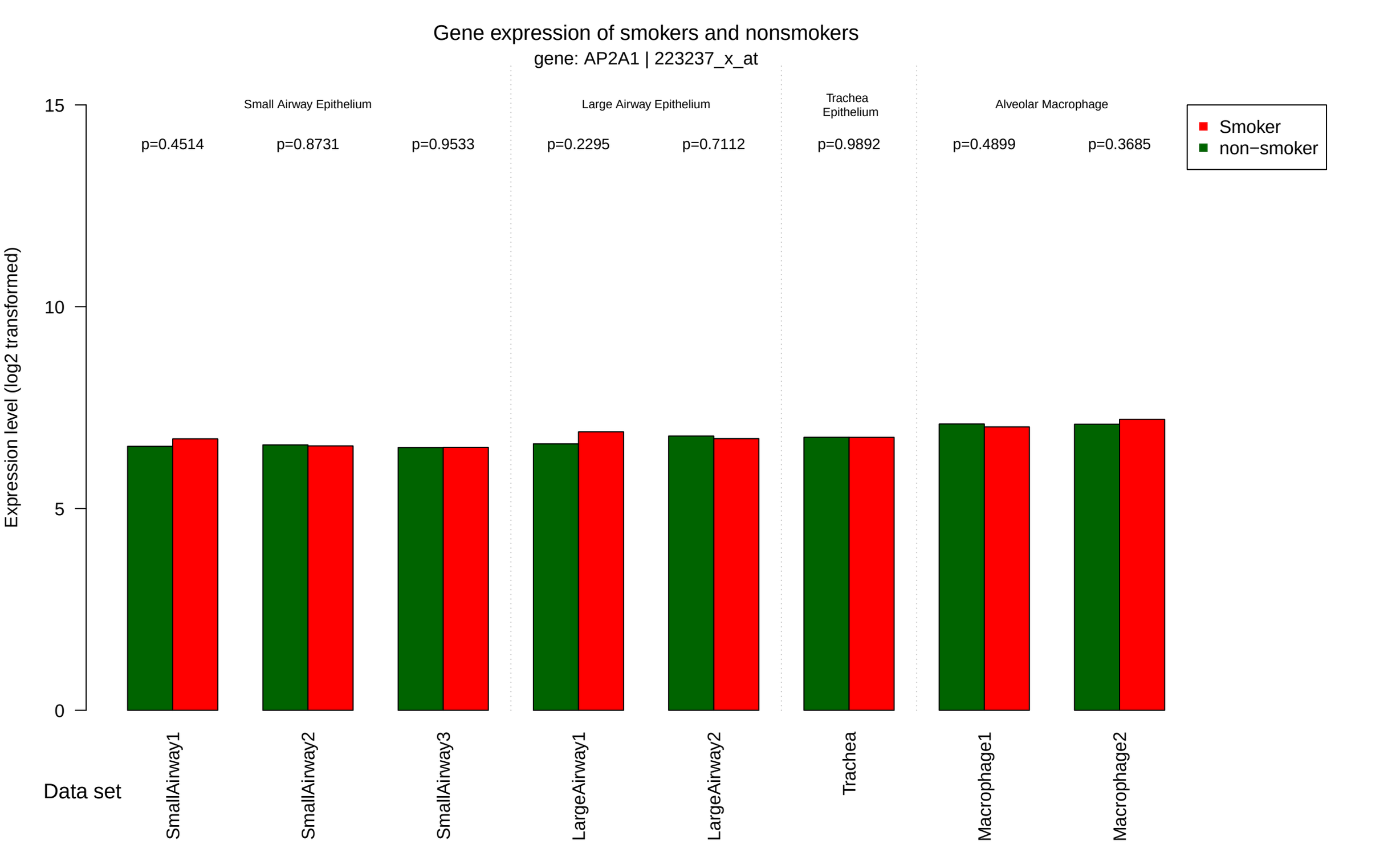
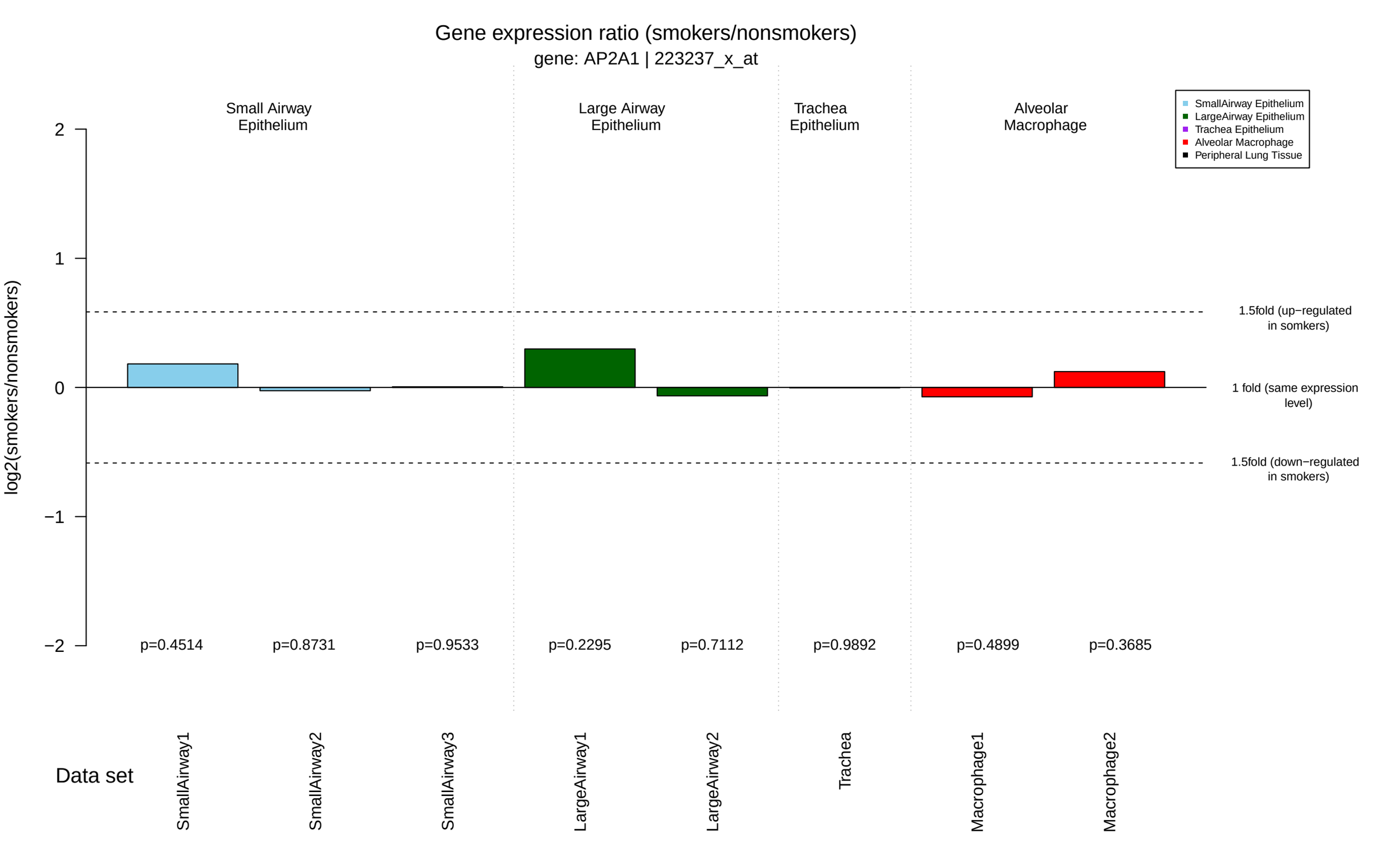
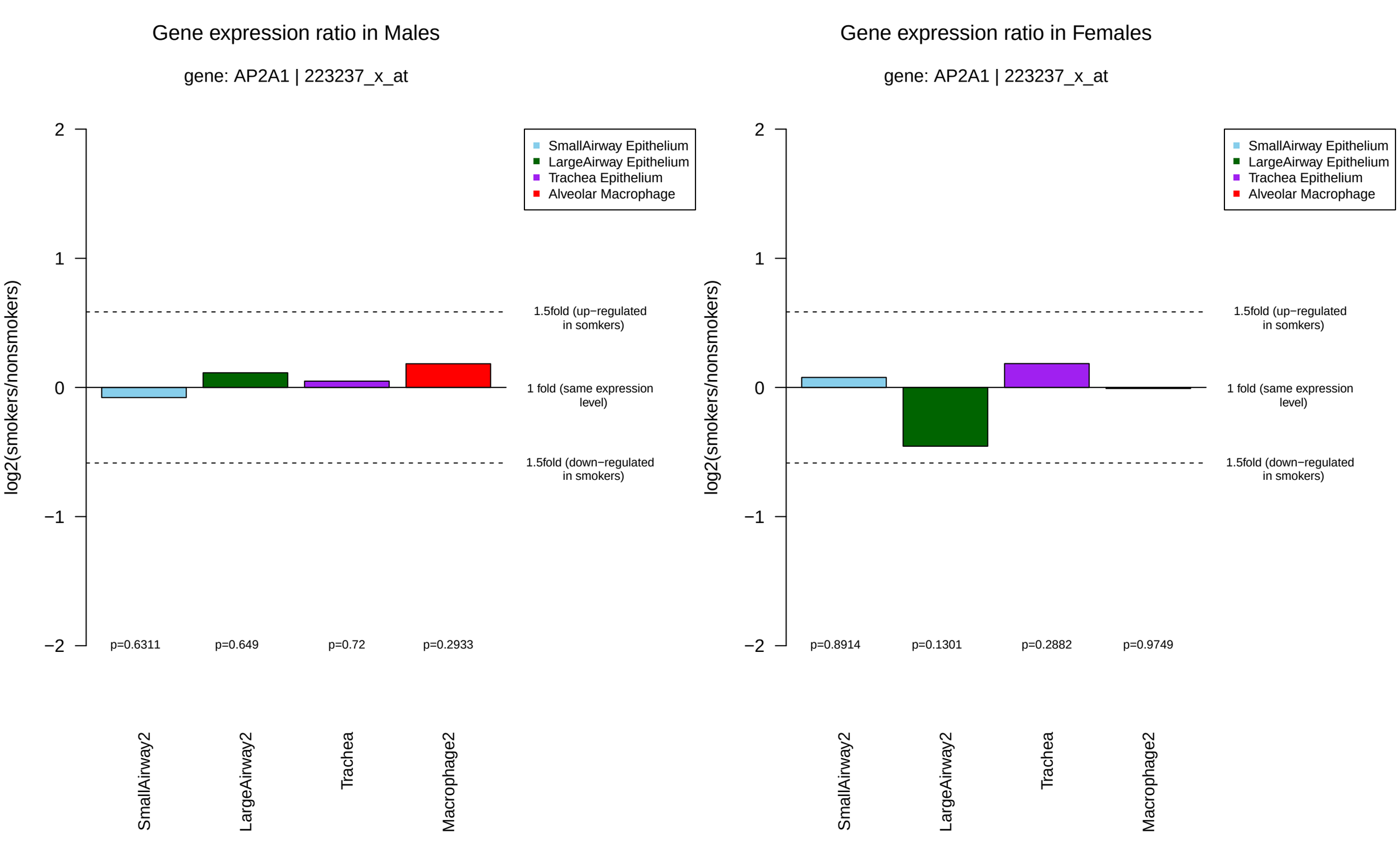
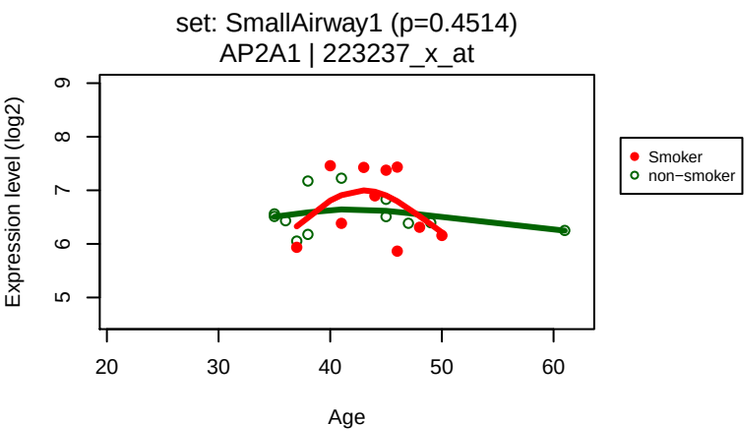
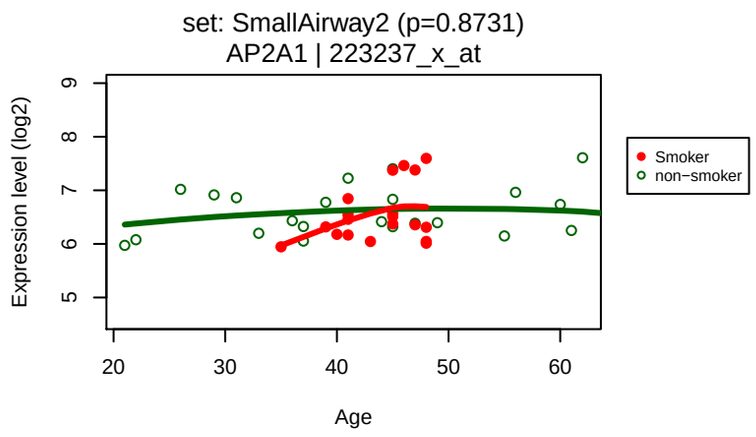
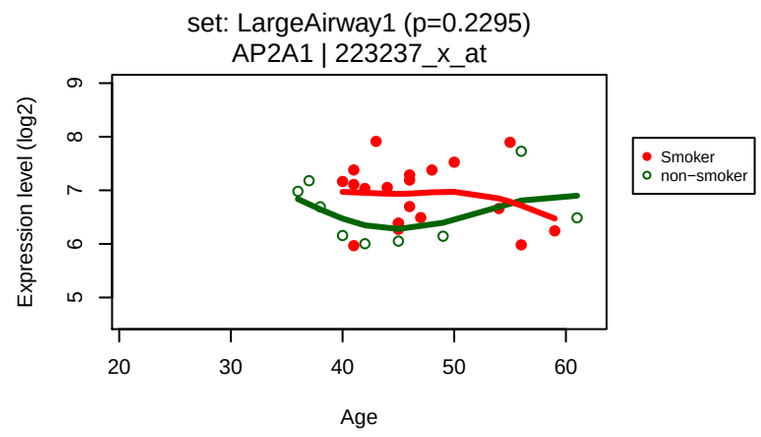

Gene expression of smokers and nonsmokers
gene: AP2A1 | 223237_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4514
p=0.8731
p=0.9533
p=0.2295
p=0.7112
p=0.9892
p=0.4899
p=0.3685
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

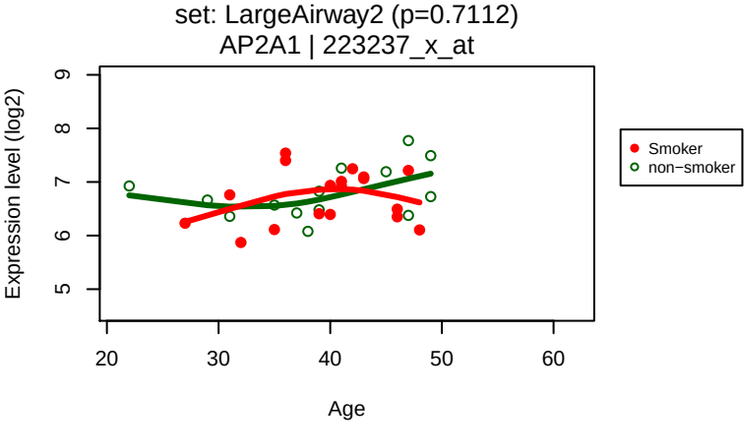

Gene expression ratio (smokers/nonsmokers)
gene: AP2A1 | 223237_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4514
p=0.8731
p=0.9533
p=0.2295
p=0.7112
p=0.9892
p=0.4899
p=0.3685
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

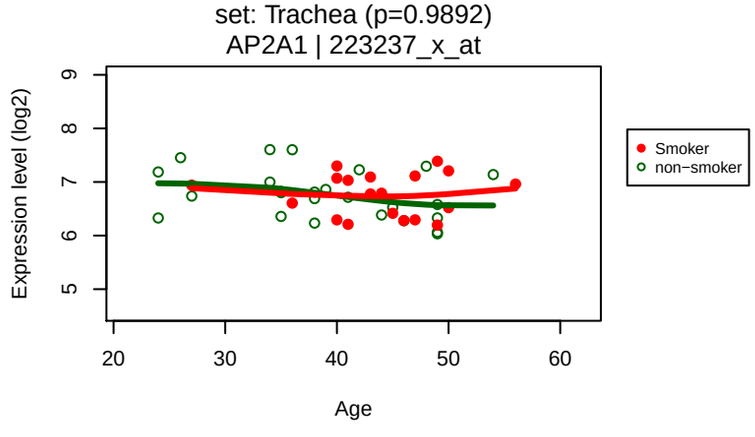

Gene expression ratio in Males
gene: AP2A1 | 223237_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6311
p=0.649
p=0.72
p=0.2933
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AP2A1 | 223237_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8914
p=0.1301
p=0.2882
p=0.9749
SmallAirway2
LargeAirway2
Trachea
Macrophage2

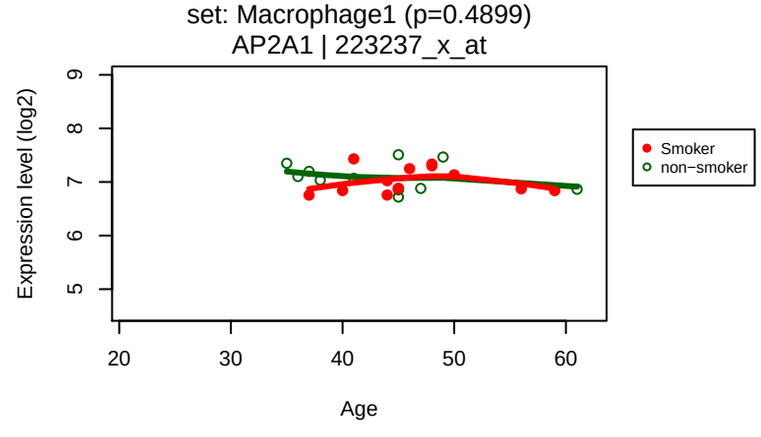

set: SmallAirway1 (p=0.4514)
AP2A1 | 223237_x_at
Expression level (log2)
Age
Smoker
non−smoker

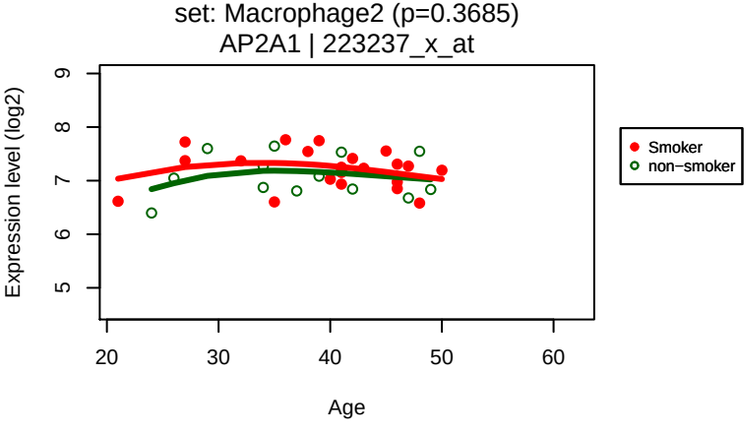

set: SmallAirway2 (p=0.8731)
AP2A1 | 223237_x_at
Expression level (log2)
Age
Smoker
non−smoker

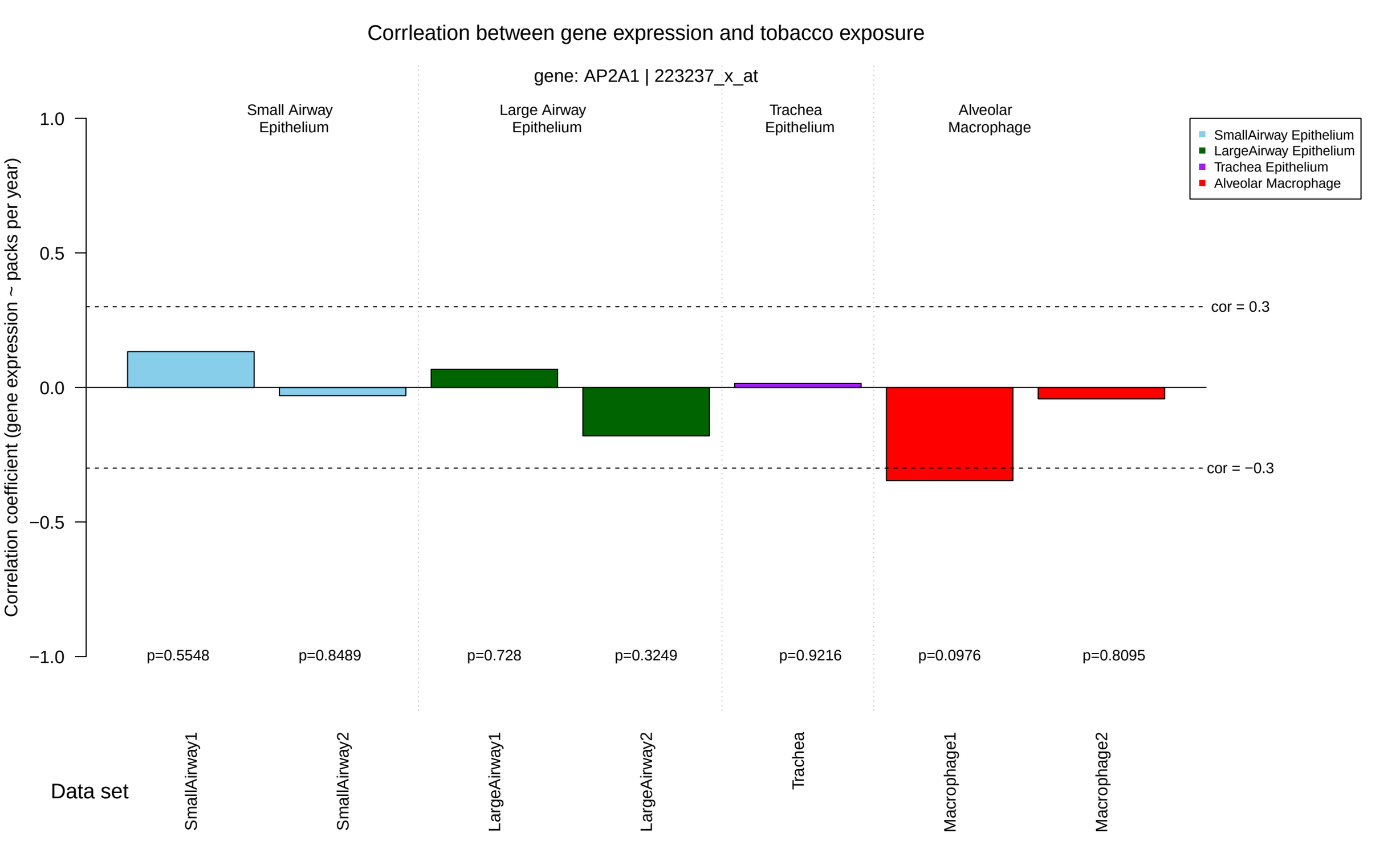

set: LargeAirway1 (p=0.2295)
AP2A1 | 223237_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7112)
AP2A1 | 223237_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9892)
AP2A1 | 223237_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4899)
AP2A1 | 223237_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3685)
AP2A1 | 223237_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AP2A1 | 223237_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5548
p=0.8489
p=0.728
p=0.3249
p=0.9216
p=0.0976
p=0.8095
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2